

Lecture 1:
Course Introduction & The Economic Approach (I)
ECON7720: Ecological and Environmental Economics

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Semester 2, 2026 | Required reading: Tietenberg & Lewis, Ch. 1 & 2

Course Information

Contact

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Consult hours: Fridays 10–11am (from Week 2), Hartley Teakle Building (83), Rm C413.

Classes

- **Lectures:** Fridays 8–10am, Hartley Teakle Building (83), Rm C416.
- **Tutorials:** Fridays 11am–12pm, Michie Building (09), Rm 216.
(weekly from Week 2, combining *problem solving* with *discussion questions*)

Part A: Course Administration

Welcome to ECON7720

- **Goal:** learn to think about environmental issues as an economist.
- **Tools:** we will use graphical analysis and some algebra, and engage with research.
- **Assumed background:** applied microeconomics (demand/supply, marginal analysis).

Two traditions, one course

- *Environmental economics* (neoclassical: internalising externalities).
- *Ecological economics* (biophysical limits, sustainability).
- Like the textbook, we take a pragmatic approach and draw on both.

Student Enquiries — School of Economics

Always check the **Course Profile (ECP)** & **Blackboard** first.

Academic / content → Course Coordinator (me)

Assessment format and content; learning materials; marks and assessment feedback; anything course-content related.

Admin → Professional Staff Team (enquiries@economics.uq.edu.au)

Timetable and tutorial allocation; Blackboard access; extension requests; deferred / supplementary exams. *Remember to include the course code in the subject line.*

General Course Information

- **Course description.** The economic analysis of environmental problems: *market failure and externalities*; *pollution-control* instruments (taxes, tradable permits, standards, liability); *climate change*; *valuing the environment* and *cost-benefit analysis*; *ecosystem services*; and *sustainable development*.
- **Course aims.**
 - to explain *why* environmental problems arise as failures of markets and institutions;
 - to evaluate the policy instruments available to address them;
 - to value environmental goods and appraise trade-offs;
 - to apply these tools to current environmental challenges.

Where to find things

- **Electronic Course Profile (ECP):** schedule, deadlines, policies.
- **Blackboard:** slides, assessment instructions/rubrics, announcements, the reading list.

Learning Resources

Required

Tietenberg, T. & Lewis, L., *Environmental and Natural Resource Economics*, 12th ed. (International Student Edition), Chapters 1 & 2.

Core lecture material is built around the relevant chapter. e-book via the Library, or buy a hard copy.

Recommended readings

- Costanza, R. (1989). What is ecological economics? *Ecological Economics*, 1(1), 1–7.
- Noussair, C. & van Soest, D. (2014). Economic experiments and environmental policy. *Annual Review of Resource Economics*, 6, 319–337.
- Lueck, D. (2002). The extermination and conservation of the American bison. *Journal of Legal Studies*, 31(2), S609–S652.

Lecture Schedule

	Date (Fri)	Topic	Key elements
1	31 Jul	Introduction & Economic Approach I	Efficiency, surplus, externalities, property rights
2	7 Aug	The Economic Approach II	Common-pool resources, public goods, liability
3	14 Aug	Economics of Pollution Control I	Efficient pollution level; damage vs. abatement cost
4	21 Aug	Economics of Pollution Control II	Emission taxes; tradable permits; cost-effectiveness
5	28 Aug	Air Pollution	Criteria pollutants; CAA; SO ₂ trading; transport
6	4 Sep	Water Pollution	CWA; ambient instruments; nutrient trading
7	11 Sep	Climate Change I	GHGs; impacts; SCC; discounting; Stern vs. Nordhaus
8	18 Sep	Climate Change II	Carbon pricing; offsets; international agreements
9	25 Sep	Evaluating Trade-offs	Cost-benefit analysis; discounting; uncertainty
	28 Sep–2 Oct	Mid-semester break — no class	
10	9 Oct	Valuing the Environment	Use & non-use value; stated & revealed preference
11	16 Oct	Valuation & Cost-Benefit Analysis	Applied valuation & policy appraisal in practice
12	23 Oct	Ecosystem Services	Natural capital; payments for ecosystem services
13	30 Oct	Sustainable Development	Weak vs. strong sustainability; Kuznets curve; trade

Objectives

On successful completion of this course you should be able to:

- **Demonstrate an understanding** of how economic principles explain environmental problems as **market failures**: externalities, public goods, common-pool resources, and missing/ill-defined property rights.
- **Demonstrate an understanding** of the main **policy instruments** (taxes, tradable permits, standards, liability) and their efficiency, cost-effectiveness, and distributional implications.
- **Apply** methods for **valuing environmental goods** and for **evaluating trade-offs** (cost-benefit analysis and discounting).
- **Critically analyse** contemporary challenges (climate change, ecosystem services, and sustainable development) from an economic perspective, including developing-country contexts.

Assessment

Category	Task	Weight	Due date
Paper / report	Policy Briefing: Research Proposal	10%	21 Aug 2026, 5:00 pm
Paper / report	Policy Briefing: Final Report	40%	9 Oct 2026, 5:00 pm
Examination	Final Exam (in-person, closed book)	50%	Exam period (7–21 Nov)

Assessment 1: Policy Briefing — Research Proposal (10%)

The Policy Briefing

You act as an environmental economist advising the Minister for the Environment: identify an environmental problem, diagnose its economic cause, and recommend a policy tool, drawing on the academic literature. The *Proposal* secures feedback that feeds into the *Final Report*.

Purpose

Identify an appropriate environmental problem and relevant academic resources to analyse in your Policy Briefing.

- You will receive **feedback** from the marker on your proposed environmental problem, which you can incorporate when developing your final Policy Briefing.
- Detailed instructions, marking criteria, and a template are available on the course **Blackboard** site.

Assessment 2: Policy Briefing — Final Report (40%)

The task

You are working as an environmental economist for the **Minister for the Environment**. Your task is to write a policy briefing on an environmental problem of your choice with a recommendation on how to best address the problem using environmental economics tools.

- The briefing should demonstrate your understanding of the **causes** of environmental issues from an economic perspective.
- Identify and critically analyse possible **environmental economic tools** which could be used to deal with the problem, drawing from the relevant academic literature.
- The *Research Proposal* secures early feedback that feeds directly into this report.

Assessment 3: Final Exam (50%)

- **Comprehensive** — covers all course content.
- **Question types:** multiple choice, short answer, and calculations.
- **Format:** in-person, closed book, paper-based.
- **Duration:** 120 minutes + 10 minutes planning time.
- A **sample exam** will be provided during the semester.

Part B: Course Introduction

Lecture Outline

1. What **environmental & natural-resource economics** is: scope and core questions
2. How the economy sits *inside* the environment: the **materials/energy balance**
3. The three functions the environment provides: **source, sink, amenity**
4. The scale of today's environmental problems (climate, air, water, biodiversity, plastics)
5. **Environmental** vs **ecological** economics; sustainability & intergenerational equity
6. The role of economics in policy: efficiency vs equity; positive vs normative
7. The **efficiency toolkit**: demand, supply, surplus, and the $MB = MC$ rule
8. **Dynamic efficiency**: discounting, present value, and the marginal user cost

Motivating examples

Easter Island (Rapa Nui)

- Settled by Polynesians (c. 900–1200 CE).
- Rapa Nui was once covered in palm forest.
- Wood was cleared for farming, canoes, and to move the giant *moai* statues.
- By the 1600s the forest was **gone**: no canoes for fishing, soil erosion, falling yields, population decline.

The economist's reading: an **open-access** resource exploited past its capacity to renew (Tietenberg & Lewis, Ch. 1).

Recommended podcast: [Fall of Civilizations, Ep. 6: "Easter Island — Where Giants Walked."](#)

See also: Brander & Taylor (1998), "[The Simple Economics of Easter Island](#): A Ricardo-Malthus Model of Renewable Resource Use," *The American Economic Review* 88(1), 119–138.

Figure 1: Polynesian migration to Rapa Nui



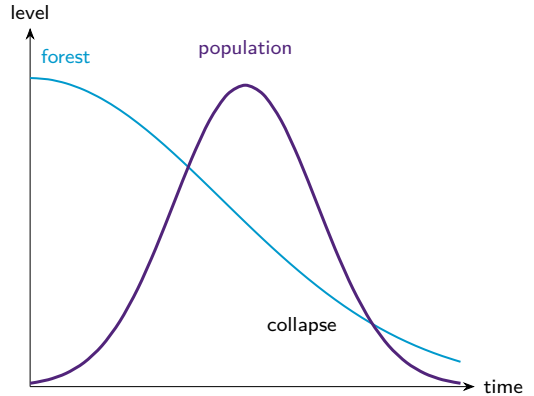
Source: Thorsby (2016), *Front. Ecol. Evol.* 4:118.

Easter Island — rise and collapse

Figure 2: Moai statues, Rapa Nui



Photo: Thor Heyerdahl / Kon-Tiki Museum.



Rise and collapse
population grows on a limited resource.

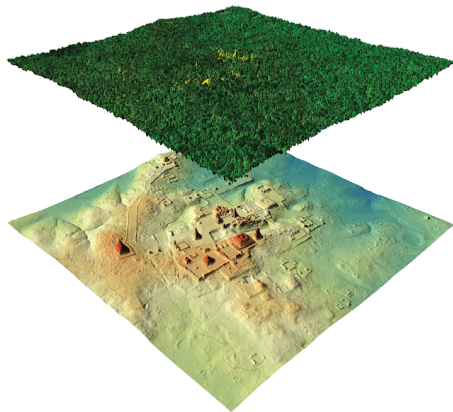
The collapse of the Classic Maya

Across the southern lowlands the Classic Maya flourished for centuries, then cities were abandoned around 800–900 CE.

- Leading factors: population pressure, intensive agriculture, **deforestation** and soil degradation, made fatal by repeated multi-decadal **droughts**.
- Collapse was multi-causal and is still debated, but environment–economy feedbacks are central to every account.

Recommended podcast: [Fall of Civilizations](#), Ep. 1: “The Maya — Forests of Kings and Kingdoms.”

Figure: Maya city revealed by LiDAR



Source: Canuto et al. (2018), *Science*; [LiDAR Magazine](#).

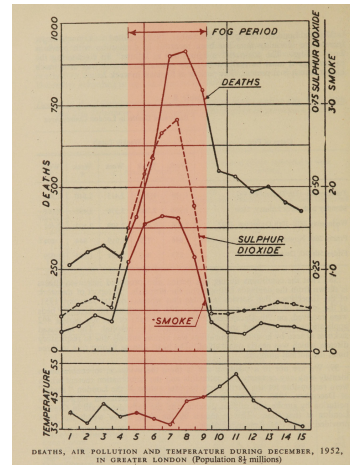
The Great Smog of London, December 1952

A period of unusually cold weather and a windless temperature inversion trapped coal smoke over London for five days (5–9 December 1952).

- An estimated 4,000 excess deaths at the time; later analyses put the toll as high as 12,000.
- Burning coal was cheap for each household and firm, but its cost in damaged health fell on everyone.
- The shock led directly to the UK Clean Air Act 1956.

Recommended: [von Hinke & Sørensen \(2023\)](#), "The Long-Term Effects of Early-Life Pollution Exposure: Evidence from the London Smog," *Journal of Health Economics*, 92, 102827.

Figure 3: Deaths, pollution & temperature



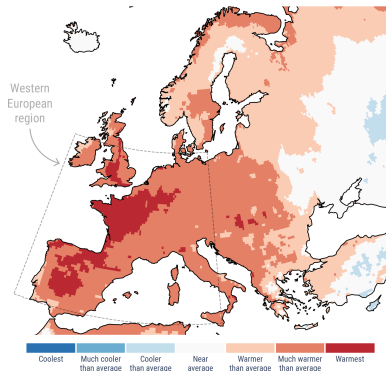
Source: UK Ministry of Health (1954),
Mortality and Morbidity during the

London Fog of December 1952.

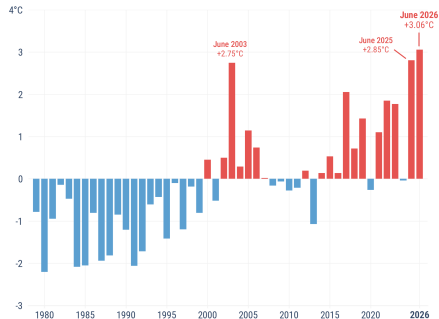
Heatwaves, June 2026

June 2026: warmest June on record for Western Europe

Anomalies and extremes in surface air temperature for June 2026



Surface air temperature anomalies in June averaged over western Europe



Source: Copernicus Climate Change Service (C3S/ECMWF), July 2026.

- Over 10,000 excess deaths across Europe during June 2026; >9,000 among those aged 65+.

What these cases teach us

In each, the **environment underpins the economy**, and harm followed from **failures of markets and institutions**: open access, missing prices, no account of the future.

The questions that emerge

- *Why* did rational individuals collectively destroy what they depended on?
- What **prices, property rights, or policies** were missing?
- How would we **value** the lost forest, clean air, or stable climate, and weigh the trade-offs?

These are the questions this course gives you the tools to answer.

What is environmental & natural-resource economics?

The application of **economic principles** (scarcity, incentives, trade-offs, and equilibrium) to the use of the natural environment and to natural resources.

- **Natural-resource economics**: how we extract and allocate resources *over time* (minerals, fisheries, forests, water); the *source* side.
- **Environmental economics**: how production and consumption damage the environment, and how to manage *pollution and residuals*; the *sink* side.

The two are inseparable

Both rest on a single idea: the environment is a **scarce asset** that yields a flow of services, and markets routinely **misprice** it.

Some core questions

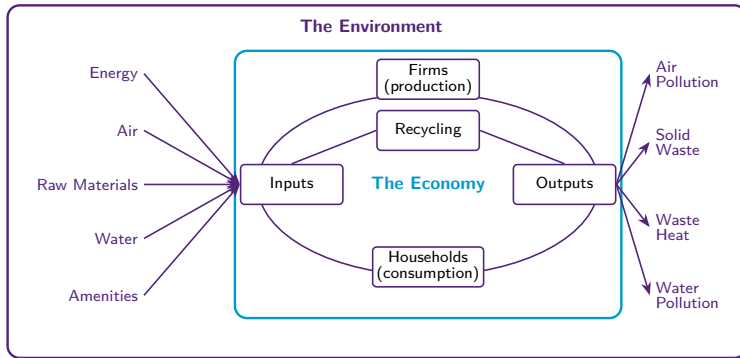
Recurring questions in this course

1. How much pollution / extraction is “too much”? (the efficiency question)
2. Why do unregulated markets get it wrong? (externalities, public goods, missing markets)
3. Which policy fixes it best: taxes, permits, standards, property rights? (instrument choice)
4. Who gains and who loses, today and across generations? (equity & sustainability)

The economy is a subsystem of the environment

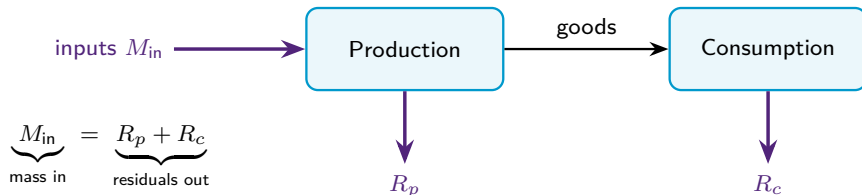
The economy is not a closed circular flow. It draws **resources and energy** from the environment and returns **residuals** to it.

Figure 4: Materials balance model



Source: Tietenberg & Lewis, *Environmental & Natural Resource Economics*, 12th ed., Figure 2.1.

Materials and energy balance



- “Goods” are temporary: eventually they too become residuals.
- You cannot “throw away” matter, only **relocate** or **transform** it.
- Recycling reduces M_{in} but never to zero.
- More broadly, **reuse**, **repurposing**, and **remanufacturing** also reduce virgin material intake.

The laws of thermodynamics

First law: conservation of mass/energy

Matter and energy are **neither created nor destroyed**. Hence the mass-balance identity: *what goes in must come out as residuals*.

Second law: entropy

Every transformation **degrades** energy quality: usable (low-entropy) inputs become unusable (high-entropy) waste heat. Recycling and abatement are **never free**; they cost energy.

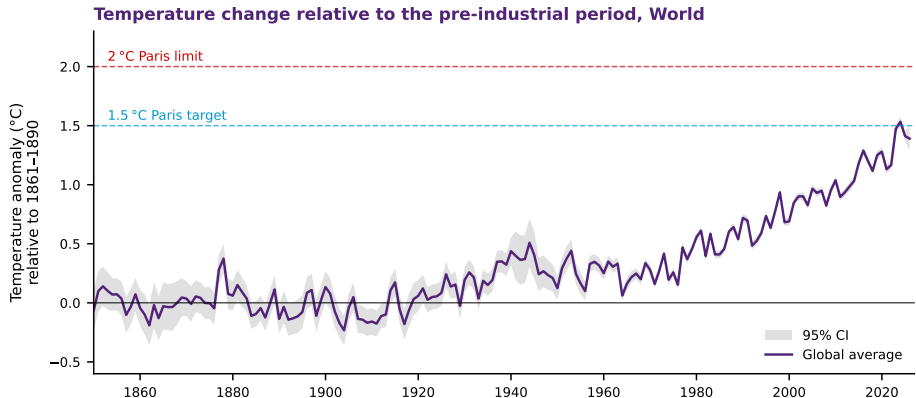
These laws bound what is physically possible: **100% recycling is impossible**, and growth in throughput cannot continue without limit on a finite planet.

Three functions the environment provides

1. **Source**: supplier of raw materials and energy (minerals, fuels, timber, fish, water).
2. **Sink**: receptacle that **assimilates** waste and residuals (CO_2 , effluent, tailings) up to a finite capacity.
3. **Amenity**: climate regulation, clean air and water, biodiversity, recreation, spiritual and cultural value.

The three functions **compete**. Using the river as a sink for effluent degrades its amenity (swimming, fishing) and its source value (drinking water). This rivalry is the root of environmental conflict.

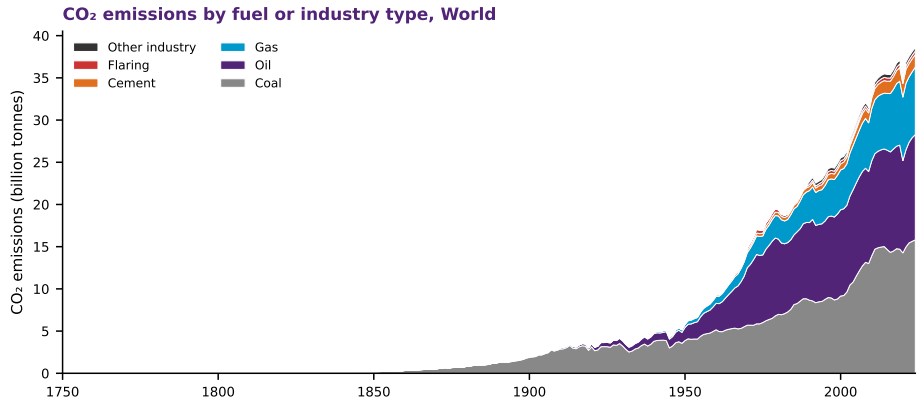
Scope & magnitude — global warming



Source: Met Office Hadley Centre – HadCRUT5 (2026); Our World in Data | CC BY. *Note:* Wider CIs in early decades reflect sparser station and ocean coverage.

Global mean surface temperature: $\approx +1.4^{\circ}\text{C}$ above the 1861–1890 baseline. Paris Agreement targets “well below 2°C ”, pursuing 1.5°C .

Scope & magnitude — CO₂ emissions



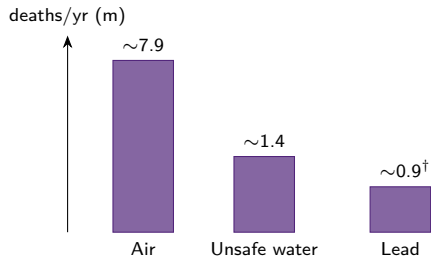
Source: Global Carbon Budget (2025); Our World in Data | CC BY.

Atmospheric CO₂: ~280 ppm pre-industrial → **>420 ppm** today, the highest in >800,000 years (IPCC AR6 WG1, 2021; ice-core record: Lüthi et al., 2008, *Nature*). *ppm* = *parts per million*: the number of CO₂ molecules per million molecules of air.

Scope & magnitude — air pollution & health

The largest environmental health burden

- Air pollution causes an estimated 6.7 million (WHO, 2024) to 7.9 million (State of Global Air, 2025) premature deaths per year.
- Around 99% of the world's population breathes air exceeding the WHO PM_{2.5} guideline of 5 $\mu\text{g}/\text{m}^3$ (revised downward in 2021; WHO, 2022).
- Fine particulates (PM_{2.5}) drive heart and lung disease, stroke and cancer (WHO, 2024).

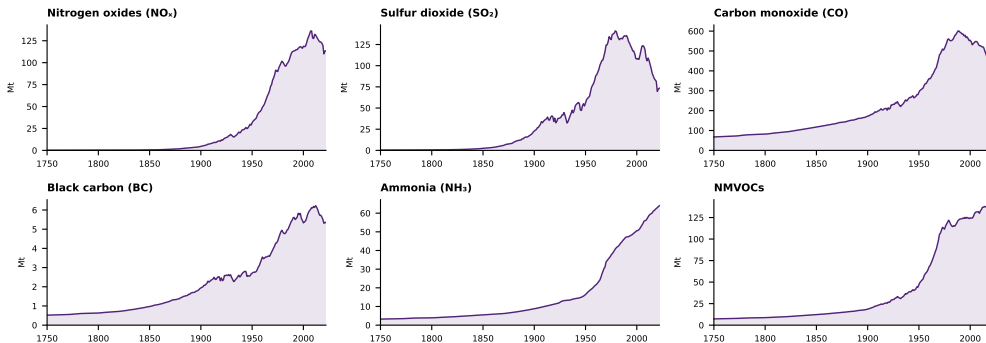


Indicative global mortality estimates. Air: State of Global Air (2025); Water: WHO (2023); Lead: GBD/IHME (2019), primarily cardiovascular deaths from lead-induced hypertension. [†] Larsen & Sánchez-Triana (2023, *Lancet Planetary Health*) estimate 5.5m lead-related cardiovascular deaths using broader dose-response pathways beyond hypertension alone.

- PM_{2.5} = particulate matter with diameter ≤ 2.5 micrometres (μm), $\sim 1/30$ th of a human hair; the “2.5” refers to *particle size*. The WHO guideline of 5 $\mu\text{g}/\text{m}^3$ is the safe *concentration* of these particles in air.
- PM_{2.5} is especially dangerous because particles this small penetrate deep into the lungs (alveoli) and cross into the bloodstream. Coarser particles (PM₁₀, ≤ 10 μm) are mostly trapped in the nose and throat.

Scope & magnitude — air pollutant emissions

Emissions of air pollutants, World, 1750-2022

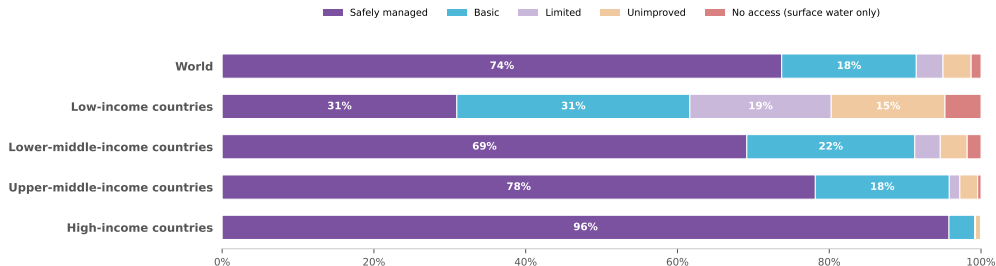


Source: Hoesly et al. (2024) – Community Emissions Data System (CEDS); Our World in Data | CC BY.

SO₂, CO, and NO_x peaked and declined in rich countries (Clean Air Acts, fuel switching), but NH₃ continues to rise globally, driven by agriculture in developing economies. Black carbon (BC), soot from incomplete combustion is a major component of PM_{2.5}. Non-Methane Volatile Organic Compounds (NMVOCs) react with NO_x in sunlight to form ground-level ozone.

Scope & magnitude — water

Access to drinking water by income group, 2024

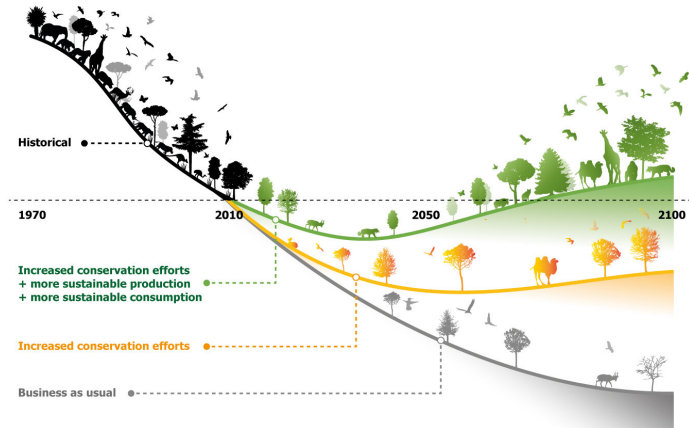


Source: WHO/UNICEF JMP (2025); Our World in Data | CC BY.

- 2.1 billion people lack safely managed drinking water (WHO/UNICEF JMP, 2025).
- Agriculture accounts for ~70% of global freshwater withdrawals.
- 25 countries (one-quarter of world population) face extremely high water stress, using >80% of renewable supply annually (WRI Aqueduct).

Scope & magnitude — biodiversity

Bending the curve of terrestrial biodiversity loss



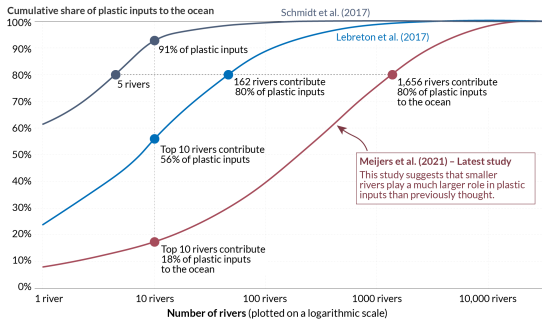
This artwork illustrates the main findings of the article, but does not intend to accurately represent its results (<https://doi.org/10.1038/941586-020-2705-y>)

Source: Leclère et al. (2020), *Nature*, 585, 551–556.

- Living Planet Index: wildlife populations down **73%** since 1970 (WWF, 2024).
- **1 million** species (animal and plant) face extinction risk (IPBES, 2019).
- Five drivers: **land-use change**, overexploitation, climate change, pollution, invasive species.
- Conservation works: targeted populations **increased 40%** (WWF, 2024).

Scope & magnitude — plastics

How many rivers are responsible for ocean plastic inputs?



Source: Meijer et al. (2021), *Science Advances*; Our World in Data | CC BY.

- Global plastic production: **460 Mt** in 2019; cumulative **>8.3 Bt** since 1950 (OECD, 2022; Geyer et al., 2017).
- 99%** from fossil fuels; largest driver of future oil demand (IEA, 2018).
- Only **~9%** ever recycled; 12% incinerated; **79%** landfilled or leaked to the environment (Geyer et al., 2017).
- 19–23 Mt/yr** enters aquatic ecosystems — rising to 53 Mt/yr by 2030 (Borrelle et al., 2020).
- Microplastics in human **blood**, **lungs**, and **placenta**.
- UN Plastics Treaty: **no agreement** as of July 2026.

The just transition challenge

Who bears the cost of environmental policy?

- Climate damages fall **disproportionately** on poor nations (as a share of GDP) and on poor households (as a share of income).
- The shift away from fossil fuels creates losers too: coal-mining regions, fossil-fuel workers, energy-intensive industries.
- A **just transition** asks whether policies can be designed so that the burdens of decarbonisation are not borne disproportionately by the most **vulnerable**.

Efficiency tells us *how big* the pie is; it is silent on *who gets what*. Transition policies that ignore distribution may be efficient on paper but politically unsustainable and ethically questionable.

The policy context — international cooperation

Global problems need coordinated responses

- Climate change, biodiversity loss and ocean pollution cross borders.
- Countries sitting at the negotiating table face **very different stakes**: low-lying island nations risk submersion; fossil-fuel exporters risk stranded assets.
- Industrialised countries imposing stringent policies may see **competitiveness** concerns (carbon leakage), while gaining new industries (renewables, clean tech).
- Developing countries may resist costly commitments without **financial and technology transfers**.

How will societies respond? Feedback loops

Two types of dynamic response

- **Positive feedback** (self-reinforcing): the process amplifies itself.
 - Warming melts permafrost → releases methane → more warming.
 - Capital accumulation: investment → output → profit → more investment.
- **Negative feedback** (self-limiting): the process corrects itself.
 - Rising scarcity → higher prices → conservation and substitution → reduced pressure.
 - Regulation response: visible damage → public pressure → policy → reduced pollution.
 - Lovelock's **Gaia hypothesis**: Earth's biophysical systems generate stabilising feedbacks that keep conditions within a habitable range.

The central question

Is the economy–environment interaction dominated by **positive** loops (self-destruction, as on Easter Island) or **negative** loops (self-correction, as after the Great Smog)? Much of this course is about designing institutions that **strengthen the negative feedbacks**.

Two traditions — environmental vs ecological economics

Environmental economics

- Rooted in **neoclassical** theory; the environment is one asset among many and problems are **market failures**.
- Tool: marginal analysis and **optimisation** of net benefits, an “efficient” level of pollution.
- Resources largely **substitutable** by capital/technology $\Rightarrow$ **weak sustainability**.

Ecological economics

- Economy as a **subsystem** of a finite biosphere; emphasises physical limits, thermodynamics, **scale**.
- Tool: **safe minimum standards** / precaution rather than fine optimisation.
- Natural capital is **complementary**, often **non-substitutable** $\Rightarrow$ **strong sustainability**.

This course is grounded in environmental economics but takes ecological critiques seriously.

Preview — weak vs strong sustainability

The substitutability question

Total capital = manufactured capital K_m + natural capital K_n .

- **Weak sustainability:** keep *total* capital non-declining; K_m may substitute for K_n (drawdown of resources is fine if we build enough machines/knowledge).
- **Strong sustainability:** keep *natural* capital itself non-declining; some natural assets (climate, biodiversity) are **critical** and cannot be replaced.

Sustainability & intergenerational equity

Development that “meets the needs of the present without compromising the ability of **future generations** to meet their own needs” (Brundtland, 1987).

- Raises an **intergenerational** fairness question markets handle poorly: the unborn cannot bid.
- Central analytical lever: the **discount rate**. A high rate weights the present heavily; a low rate protects the future.
- Efficiency alone is silent on *which generation* bears costs. That is an equity (normative) choice.

The role of economics in policy

What economics contributes

- A consistent framework to weigh costs against benefits of environmental action.
- Tools to find least-cost ways to hit a target (e.g. taxes, tradable permits, standards).
- Estimates of non-market values (e.g. clean air) so they enter decisions.

Efficiency vs equity

- **Efficiency**: is the “pie” (net social benefit) as large as possible?
- **Equity**: how is the pie (and the costs of policy) distributed across people, regions, generations?
- A policy can be efficient yet widely seen as unfair; both matter for political viability.

Positive vs normative; the use of models

Two kinds of statement

- **Positive:** what *is* / will happen; testable. “A carbon price of \$50/t cuts emissions by $X\%$.”
- **Normative:** what *ought* to be. “We *should* protect the reef even at high cost.”

Why we build models

- Models **simplify** to isolate the key mechanism.
- Good for clear, transparent assumptions and comparative-static predictions.
- Caution: conclusions are only as good as the assumptions.

Recap & bridge to “The Economic Approach”

What we established

- The economy is embedded in a finite environment governed by mass balance and entropy.
- The environment serves as **source**, **sink** and **amenity**, and these uses compete.
- Two traditions, two views on substitutability and sustainability.
- Policy needs both **efficiency** and **equity**; keep positive and normative claims distinct.

Next

Chapter 2, **The Economic Approach**: property rights, externalities, efficiency, and *why* unregulated markets fail to protect the environment.

Part C: The Economic Approach (I)

Benefits and costs

The central question

Should we undertake an action (a policy, a project, an extra unit of output)? The economic answer compares what we **gain** against what we **give up**.

$$\text{Net Benefits} = \text{Benefits} - \text{Costs}.$$

- **Benefits**: the value people place on the outcome, measured by their **willingness to pay**.
- **Costs**: the value of the resources used up, measured by their **opportunity cost**.
- **Decision rule**: act when net benefits are positive; choose the scale that **maximises** net benefits.
- At the optimum, the **marginal benefit** of the last unit equals its **marginal cost** ($MB = MC$).

The rest of this part builds the tools to measure benefits and costs.

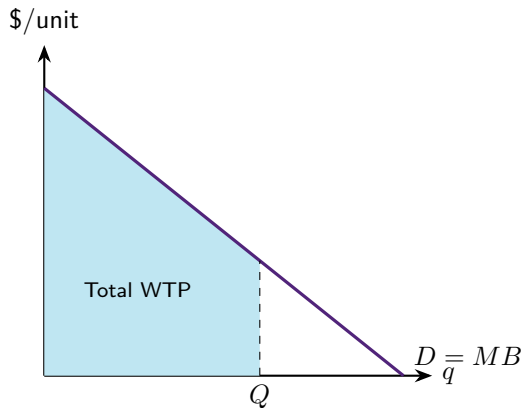
Willingness to pay and the demand curve

A person's **willingness to pay** (WTP) for a unit of a good is the maximum amount they would sacrifice to obtain it. WTP reveals how much the good is worth *to them*.

- Array units from most-valued to least-valued. The height of the **demand curve** at quantity q is the WTP for the q th unit, the **marginal WTP**.
- Demand slopes down: as more is consumed, the WTP for additional units falls (**diminishing marginal value**).
- Hence the demand curve *is* the marginal benefit curve: $P = MB(q)$.

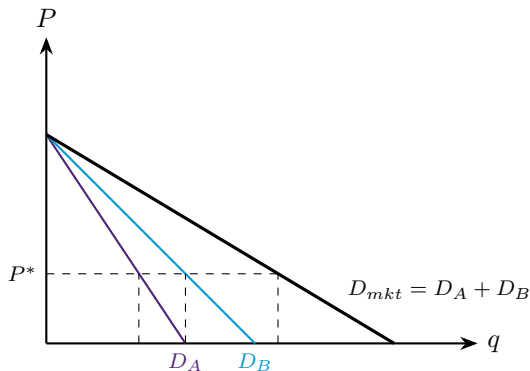
Total WTP for Q units $= \int_0^Q MB(q) dq =$ area under the demand curve up to Q .

Marginal WTP = demand; Total WTP = area beneath



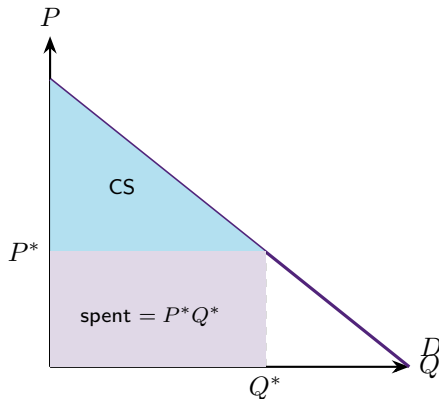
- Each thin vertical strip has width dq and height $MB(q)$: the value of one more unit.
- Summing all strips up to Q gives the **total willingness to pay**, the whole shaded area.
- This is the **gross benefit** of consuming Q units, *before* subtracting what was paid.

Individual vs. market demand — horizontal summation



- At a given price P^* , read off each person's quantity demanded.
- Market quantity = sum of individual quantities: $Q(P^*) = q_A(P^*) + q_B(P^*)$.
- Repeat at every price $\Rightarrow$ **horizontal summation** of the individual curves.

Consumer surplus



Value received minus value paid

$$CS = \underbrace{\int_0^{Q^*} MB(q) dq}_{\text{total WTP}} - \underbrace{P^* Q^*}_{\text{expenditure}} .$$

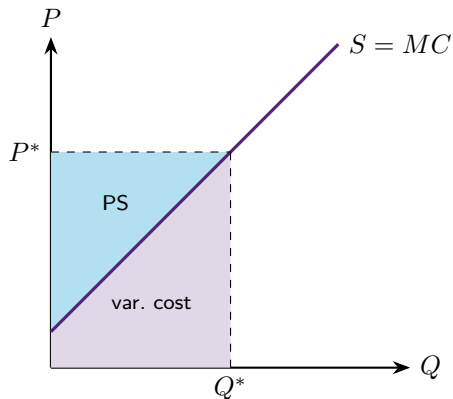
- Each buyer pays the same P^* but valued early units far above it.
- **Consumer surplus** = the area between the demand curve and the price line.
- It measures the net gain to consumers from being able to buy at P^* .

Supply, marginal cost and opportunity cost

The **opportunity cost** of producing a unit is the value of the resources in their next-best use. The **marginal cost** curve records the opportunity cost of each successive unit.

- Order units from cheapest to most expensive to produce (**increasing marginal cost**).
- In a competitive market the firm's supply curve *is* its marginal cost curve: $P = MC(q)$.
- **Total variable cost** of producing $Q = \int_0^Q MC(q) dq = \text{area under the supply curve}$.
- Market supply is again the **horizontal sum** of individual firm supplies.

Producer surplus



Revenue received minus cost incurred

$$PS = \underbrace{P^*Q^*}_{\text{revenue}} - \underbrace{\int_0^{Q^*} MC(q) dq}_{\text{variable cost}}.$$

- Early (cheap) units cost far less than P^* yet still sell at P^* .
- **Producer surplus** = the area between the price line and the supply curve.
- It is the producers' net gain from selling at P^* (a return to fixed factors / scarcity).
- $PS \neq \text{profit}$.

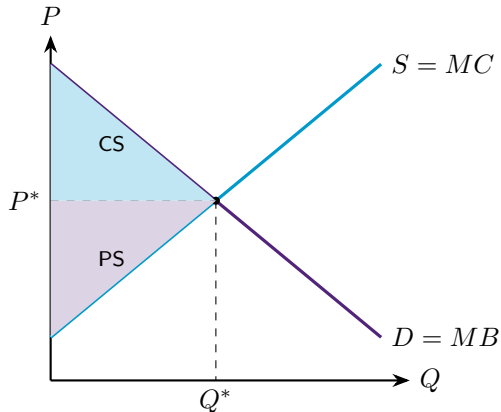
Producer surplus, profit and scarcity rent

- **Short run:** $PS = \text{Profit} + \text{Fixed Cost} + \text{Rent}$.
(Area under P^* and above MC includes the return to fixed inputs and to scarce inputs.)
- **Long run:** $PS = \text{Profit} + \text{Rent}$.
In the long run all inputs are variable (no fixed costs), so only **rent** remains = the return to **genuinely scarce inputs** (fertile land, a mineral deposit, a prime location).
- Under perfect competition, long-run economic profit $\rightarrow 0$, so $PS = \text{Rent}$.

Scarcity rent

- **Scarcity rent** is the producer surplus that **persists** in long-run competitive equilibrium. It cannot be competed away because the underlying input is genuinely scarce.
- For natural resources (oil, minerals, water rights), scarcity rent reflects the **asset value** of the stock. It is the return that makes the owner willing to hold the resource rather than sell it today.

Total surplus = net benefits to society

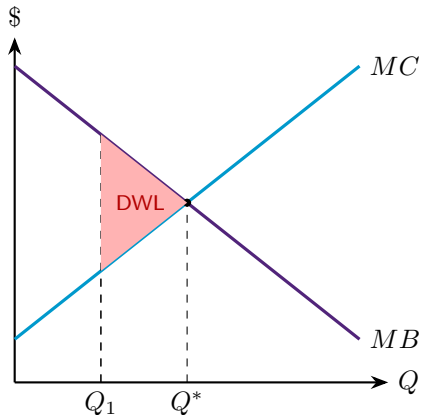


Adding the two surpluses

Total Surplus = $CS + PS$ = Net Benefits.

- Equivalently: total WTP – total cost
$$= \int_0^Q [MB(q) - MC(q)] dq.$$
- Geometrically: the whole area between the demand and supply curves up to Q .
- Society's net gain from the market is the sum of consumer and producer gains.

The efficient quantity and deadweight loss



Maximise net benefits at $MB = MC$

- For each unit up to Q^* , $MB > MC$: producing it **adds** net benefit.
- Beyond Q^* , $MC > MB$: producing it **subtracts** net benefit.
- So net benefits peak where $MB = MC$, the **efficient quantity** Q^* .
- Any $Q_1 \neq Q^*$ forgoes the shaded **deadweight loss**: value that could have been created but wasn't (under-production here).

The static efficiency criterion & the equimarginal principle

Static efficiency (formal statement)

An allocation of resources is **statically efficient** if it maximises the net benefits (total surplus) from all uses of those resources in a given period. Equivalently, no reallocation can raise total surplus; it is **Pareto efficient**.

First-order condition. Maximising $NB(Q) = \int_0^Q [MB - MC] dq$ gives

$$\frac{dNB}{dQ} = MB(Q^*) - MC(Q^*) = 0 \implies MB(Q^*) = MC(Q^*).$$

Equimarginal principle (allocation across uses)

To allocate a fixed resource efficiently among several uses, set the **marginal net benefit equal across all uses**: $MB_1 - MC_1 = MB_2 - MC_2 = \dots$. If marginals differed, shifting a unit from the low- to the high-marginal use would raise total net benefits.

Why this matters for environmental economics

- The competitive equilibrium Q^* where $MB = MC$ is the **efficiency benchmark**.
- Environmental problems arise when **private** marginal cost $\neq$ **social** marginal cost (pollution externalities), so the market quantity is *not* efficient.
- The surplus toolkit lets us **measure** that inefficiency (a deadweight loss) and value corrective policies.
- But many environmental decisions span **time**: climate, depletable resources. We need to compare benefits and costs across time.

$\Rightarrow$ from static to *dynamic* efficiency.

Dynamic efficiency — present value and discounting

A dollar tomorrow is worth less than a dollar today

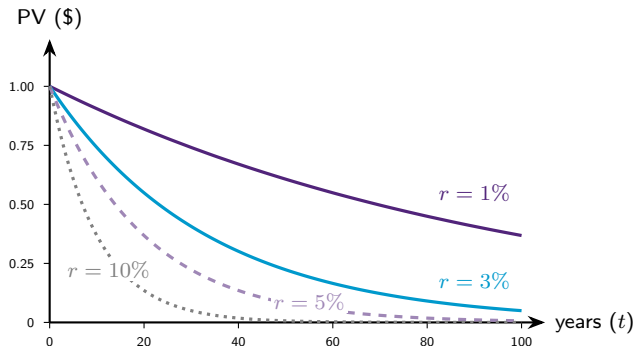
- **Why discount?** Opportunity cost of capital (\$1 invested grows), pure time preference (impatience), and uncertainty/risk.
- **Present value** converts a future amount B_t into today's equivalent using discount rate r :

$$PV(B_t) = \frac{B_t}{(1+r)^t}, \quad PV \text{ of a stream} = \sum_{t=0}^T \frac{B_t}{(1+r)^t}.$$

- **Net present value** of a project: $NPV = \sum_{t=0}^T \frac{B_t - C_t}{(1+r)^t}$.
- The discount factor $(1+r)^{-t}$ shrinks fast: at $r = 5\%$, a payoff in 50 years is worth only ≈ 0.087 of its face value today.

How the discount rate affects present value

Present value of \$1 received in year t



Each curve shows how \$1 received in year t shrinks when discounted back to today.

- A **high** discount rate rapidly shrinks the present value of future payoffs. The future “almost doesn’t count.”
- A **low** rate keeps future values large. The future matters almost as much as today.
- For **climate policy**, where damages peak decades from now, the choice of r is decisive.

Present value under risk and uncertainty

Future benefits and costs are rarely known with certainty. Multiple outcomes are possible, each with a different probability.

Expected present value of net benefits

Weight each outcome by its probability:

$$EPVNB_j = \sum_{i=1}^I P_i \times PVNB_{ij}, \quad j = 1, \dots, J$$

where P_i is the probability of outcome i , and $PVNB_{ij}$ is the present value of net benefits for policy j under outcome i . Choose the policy with the **highest $EPVNB$** .

The discount-rate debate — Stern vs. Nordhaus

- The choice of r dominates climate change cost estimates: damages occur far in the future, so the present value is hugely sensitive to r .
- **Stern Review (2007)**: low rate ($\approx 1.4\%$), reflecting a near-zero pure rate of time preference; treats future generations almost equally. $\Rightarrow$ **act aggressively now**.
- **Nordhaus (DICE)**: higher, market-calibrated rate ($\approx 4\text{--}5\%$); defers heavy abatement, lets growth fund future action. $\Rightarrow$ **gradual ramp**.

Stern, N. (2007). *The Economics of Climate Change: The Stern Review*. Cambridge University Press.

Nordhaus, W. D. (2019). Climate change: The ultimate challenge for economics. *American Economic Review*, 109(6), 1991–2014.

The dynamic efficiency criterion

Definition

An allocation of resources **across time** is **dynamically efficient** if it maximises the **present value of net benefits** over the entire time horizon:

$$\max \sum_{t=0}^T \frac{B_t - C_t}{(1+r)^t}.$$

- Static efficiency ($MB = MC$ each period) is the special case with one period (or r irrelevant).
- Dynamic efficiency is the right benchmark for **depletable resources**: an oil barrel used today cannot be used tomorrow.
- This intertemporal trade-off creates a cost that the static framework misses: the **marginal user cost**. The marginal user cost only arises when you introduce time and a finite stock.

Extraction cost vs. marginal user cost

Marginal extraction cost

The **physical cost** of getting one more unit out of the ground: labour, equipment, energy, processing.

Marginal user cost

The **opportunity cost** of depleting a finite stock: the present value of the future net benefit forgone by extracting now.

The efficient price of a depletable resource

$$P^* = \underbrace{\text{marginal extraction cost}}_{\text{static analysis sees this}} + \underbrace{\text{marginal user cost}}_{\text{dynamic analysis adds this}}$$

For a **renewable** resource harvested sustainably, the stock replenishes and user cost is zero. For a **depletable** resource (oil, minerals), user cost is positive.

Two-period allocation — scarcity rent (Hotelling)

Allocate a fixed stock between today and tomorrow

A fixed stock $\bar{S}$ must be split: $q_1 + q_2 = \bar{S}$. Using a unit today means it is *not* available tomorrow.

- **Marginal user cost** (scarcity rent): the present value of the future net benefit forgone by extracting one more unit *now*.
- Efficient rule: allocate so that the **present value of marginal net benefit is equal across periods**:

$$[MB_1 - MC_1] = \frac{MB_2 - MC_2}{1 + r}.$$

- When the resource is **scarce**, $MB > MC$ in each period: the price exceeds extraction cost. The gap $MB - MC$ is the **scarcity rent** (marginal user cost), a premium above extraction cost that reflects the value of the depleted stock.
- If the stock were infinite, scarcity rent would be **zero**, price would just equal extraction cost (the static result). A positive rent is what makes the resource price sit above extraction cost.
- **Hotelling intuition**: for the rent to leave owners indifferent about *when* to sell, it must rise at the rate r .

The mathematics of dynamic efficiency (1/2)

A depletable resource with fixed stock $\bar{Q}$, linear inverse demand $P_t = a - bq_t$, and constant marginal extraction cost c . Allocate over n periods.

Total benefits and costs in period t

$$\text{Total benefits}_t = \int_0^{q_t} (a - bq) dq = aq_t - \frac{b}{2}q_t^2, \quad \text{Total cost}_t = cq_t.$$

The maximisation problem

Choose $q_1, \dots, q_n$ to:

$$\max \sum_{i=1}^n \frac{aq_i - \frac{b}{2}q_i^2 - cq_i}{(1+r)^{i-1}} + \lambda \left(\bar{Q} - \sum_{i=1}^n q_i \right)$$

where λ is the Lagrange multiplier on the stock constraint $\sum q_i = \bar{Q}$.

The mathematics of dynamic efficiency (2/2)

Differentiating with respect to q_i :

$$\frac{a - bq_i - c}{(1 + r)^{i-1}} = \lambda, \quad i = 1, \dots, n$$

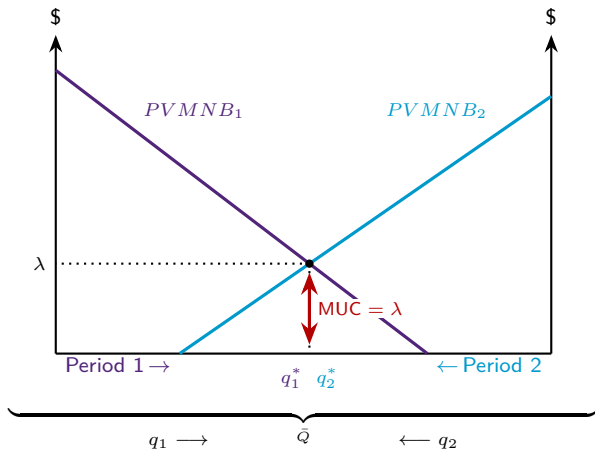
Since $(a - bq_i) = P_i$, this says:

$$\frac{P_i - c}{(1 + r)^{i-1}} = \lambda \quad \text{for all } i.$$

1. λ is the **present value of marginal user cost**, the shadow price of the stock constraint.
2. $P_i - c = \lambda(1 + r)^{i-1}$: the **marginal user cost in current value** rises at rate r over time. This is the **Hotelling rule**.
3. The complementary slackness condition $\lambda(\bar{Q} - \sum q_i) = 0$ means: if the stock is not exhausted ($\sum q_i < \bar{Q}$), then $\lambda = 0$ and $P = c$, no scarcity rent. Scarcity rent is **positive only when the constraint binds**.

Two-period dynamic efficiency — graphical intuition

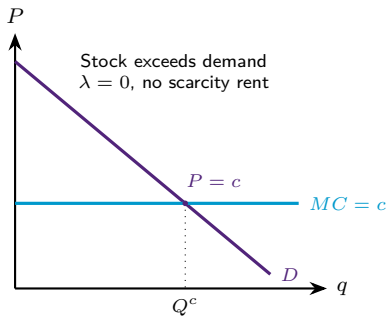
Allocating a fixed stock $\bar{Q}$ between two periods



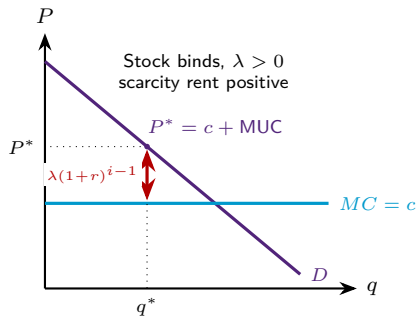
- $\bar{Q}$ is the total width. q_1 reads left-to-right; q_2 reads right-to-left.
- $PVMNB_1$ slopes down from the left axis (not discounted).
- $PVMNB_2$ slopes down from the right axis (discounted by $1/(1+r)$).
- Efficient split: where the two curves **intersect**. At that point, $PVMNB_1 = PVMNB_2 = \lambda$.
- The height λ is the **marginal user cost**: the PV shadow price of the stock.
- A higher r tilts $PVMNB_2$ down $\Rightarrow$ more to Period 1, less to Period 2.

When does scarcity rent arise? — graphical intuition

(a) Abundant stock



(b) Scarce stock



- **Panel (a):** stock is large enough to satisfy unconstrained demand (Q^c) in every period. Price = extraction cost. No scarcity rent ($\lambda = 0$).
- **Panel (b):** stock is insufficient. Each period gets less than Q^c , so $P^* > c$. The gap $P^* - c$ is the **marginal user cost** (scarcity rent): the opportunity cost of the finite stock.

Appendix

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